

FLY BY / **ONS**

When GPS dies, the drone goes blind.

We keep it flying — with passive optical terrain-matching that needs no satellite.

FIELD-TESTED

17 m median error in real flights

IN PRODUCTION

first hardware prototypes being built

PRE-SEED

\$500K · SAFE

GPS jamming makes every mission a coin flip.

Jamming is constant — and the demand is already ordered.

46,000

GNSS jamming events recorded

In conflict zones, 2024 · GPS World

+300%

YoY growth in jamming incidents

C4ADS · 2024

30,000+

Strike drones ordered, Ukraine

Zelensky / Umerov · 2025

EW jams or spoofs GPS. Drones lose coordinates, drift off course, crash, or become targets. Every platform without a satellite-independent position source is a single point of failure.

GPS-denied nav is becoming a requirement.

Capital, regulation, and demand are converging through 2026.

5%

NATO defence-GDP target by 2035

UK 2.5% by 2027 · Hague Summit 2025

2026

EU Drone Wall IOC

GPS-denied nav becomes a spec · EU programme

30K+

Strike drones ordered need GPS-denied nav

Zelensky / Umerov · 2025

ReArm Europe €800B + SAFE €150B shown as opportunity pipeline — not market size.

Optical terrain-matching replaces GPS.

Camera + onboard compute match live terrain to a pre-loaded reference map, fused with IMU.

01

Absolute position, no drift

Route-corridor map loaded pre-flight. Map-matching gives absolute coordinates — no satellite, no cumulative inertial drift.

02

Field-proven accuracy

17 m median error in real flight tests at ~200 m AGL — better than the 20-50 m the mid-strike mission requires. Down to ~3 m in best conditions.

03

Plug-in module

Feeds the autopilot over MAVLink as an external GPS source. Works with open-standard autopilots (ArduPilot / PX4) — no operator retraining.

04

Zero RF signature

Passive optical sensing only. Nothing radiates, so EW has nothing to detect, jam, or home in on.

Algorithm live and flight-tested on commodity cameras · working MVP in 6-9 months.

New test data: we beat the mission spec.

Real flights, real routes, commodity camera — verified against GPS ground truth.

17 m

Median position error

Full route, ~200 m AGL, 25 m/s cruise

~3 m

Best-condition accuracy

High-texture terrain segments

300 ms

Per-frame processing

Down from 1 s — path to real-time onboard

20–50 m

What the mission needs

Mid-strike requirement — already exceeded

WHAT THIS MEANS

Multiple test flights at different altitudes matched live video to the reference map with a 17 m median error — on a consumer-grade camera taped to the airframe. **Military mid-strike missions need 20-50 m: we already exceed the requirement, before IMU fusion tuning and model training on accumulated flight data. Target: ≤ 5 m CEP at 150-200 km.**

Module chosen — manufacturing started.

Product-form decision made: integrated hardware module. First prototype batch is in manufacturing now.

INSIDE THE MODULE

- **Optical camera** — day-mode now, night-mode on roadmap
- **Onboard compute** — runs the matching algorithm in flight
- **Encrypted map storage** — route-corridor reference maps stay inside the box — IP protected
- **MAVLink interface** — presents itself to the autopilot as an external GPS receiver
- **IMU fusion** — uses the platform IMU over MAVLink; tolerates 10–20° camera misalignment

PRODUCTION STATUS · AUG 2026

- | | |
|----------------|--|
| NOW | First prototype units in assembly — 3D-printed enclosure, branded, flight-ready |
| END AUG | 3 units built and flying at a partner test range near Lviv |
| NEXT | 10-unit prototype batch for OEM field trials on partner ISR aircraft (100 km linear routes) |
| BOM | \$1,000–1,500 per unit at prototype volumes — compute, 1 TB storage, camera, power & cooling |

Why hardware: OEMs asked for a calibrated, install-and-fly box — and onboard encrypted maps beat a web platform on security.

The approach works — map IP is the moat.

Matching algorithms will commoditise. Sovereign, fresh, GIS-grade maps will not.

17 m

ONS — own field result, Aug 2026

Median error, real flights vs GPS ground truth · target ≤ 5 m CEP

52 m

Independent benchmark (Theseus)

Median, 564 km optical-nav field test, Mar 2026 — validates the category, not our metric

Sovereign map layer

Own GIS-grade reference maps per theatre — no Maxar lock-in, available where US primes can't go

Fresh AO layer

Annual subscription keeps the area-of-operations layer current as terrain changes

Honest confidence

Position + confidence output, honest “no fix” — a slot no competitor publishes

Career-long nav team

45+ combined years in geospatial, navigation and UAV systems; national-scale cartography exit

A \$334M niche inside a \$4.8B PNT market.

≈10,000 nav units per month needed on the UA front alone — validated in 20+ OEM interviews.

\$4.8B

TAM · 2030

Assured-PNT market · 24–29% CAGR · Mordor
2025 / TBRC 2026

\$334M

SAM · 2033

GPS-denied drone-nav niche, from \$149M
(2024) · SRI 2026

\$6M

SOM · Year 1

12-mo benchmark · 2,000 units × \$3,000 ·
bottom-up, ONS est.

Demand signal from the field: OEM interviews (Jul–Aug 2026) put UA demand at ~10,000 units/month for mid- and deep-strike platforms alone — and ~300 units/month are sellable today through our existing warm OEM network.

Defence beachhead first, civil dual-use next.

Same software stack across platform classes — once MVP validates.

PRIMARY · YEARS 1-2

UA UAV OEMs

Frontline serial-production lines — 280+ leads in pipeline, first replies in

Allied defence ministries

UK MoD · USAF · allied procurement

OEM integrators

Drone primes — Rheinmetall, Northrop

EXPANSION · DRONE CLASSES · YEARS 2+

ISR drones

GPS-denied recon missions

Deep-strike UAVs

Long-range autonomous targeting

Light aviation

GNSS-loss backup navigation

Swarm formations

Coordinated multi-platform ops

Civil dual-use: Assured PNT · Logistics / BVLOS · Aviation backup · Emergency response.

Module plus recurring map subscription.

Pricing validated in 20+ OEM & operator interviews, Jul-Aug 2026.

\$3,000

Module + annual maps

Per-unit benchmark price + recurring map subscription · BOM \$1.0–1.5K leaves hardware margin plus software-grade recurring revenue

3–5 m

Premium accuracy tier

Correctors / last-mile precision for platforms where accuracy beats price — price-insensitive segment

Royalty

OEM integrators

Per-platform royalty + IP licensing for primes integrating the stack at scale

Y1 benchmark: 2,000 units × \$3,000 = \$6M. Interviews confirm a two-tier market: price-critical mass segment (mid-strike) and price-insensitive precision segment — the module + subscription model serves both.

Hardware in production, MVP in 6-9 months.

Each phase converts capital into measurable validation. Path to Seed is clear.

PHASE 1 · DONE

Core algorithm

- Map-matching algorithm live
- Field flights: 17 m median, ~3 m best
- 300 ms/frame processing

Bootstrapped

PHASE 2 · NOW

Hardware prototypes

- First units in manufacturing
- 3 units flying by end of Aug
- MAVLink external-GPS integration
- Test range secured (Lviv)

Pre-Seed \$500K

PHASE 3 · M6-M9

Working MVP

- GPS-free flight validated
- ≤ 5 m CEP target met
- 10-unit batch in OEM trials
- Night-mode prototype

+ UK DASA Open Call

PHASE 4 · M9-M12

Seed round

- LOI $\times$ 3-5 from UA OEMs
- DASA / DIANA decision
- Seed raise launched

Seed \$3M-\$3.5M target

Three founders. Five years together.

45+ combined years in geospatial, navigation, and UAV systems.



20+ YEARS GEOSPATIAL

Eugene

CEO · Co-Founder

Built a national-scale cartography company 2016–2023. Founded GIS-POINT (UK), maker of FLY BY.



FULL-STACK ENGINEERING

Andrew

CTO · Co-Founder

10+ years scalable software systems. Owner of ONS algorithms — map-matching, IMU fusion, autopilot integration.



30 YEARS · PRIOR EXIT

Dr. Alex

Chief Scientific Advisor · Co-Founder

PhD, 30+ years navigation & UAV systems. Built and exited a prior UAV company. Chair, All-Ukrainian AeroGeodetic Association.

Together since 2021 — the team that builds the maps now builds the navigation. Lean, capital-efficient.

Pre-Seed: \$500K from prototype to MVP.

SAFE · \$2.25M valuation cap · 20% discount · founders co-investing alongside.

USE OF FUNDS



MOMENTUM & NON-DILUTIVE PIPELINE

- 41** accelerator & VC applications submitted (US, EU, UA) — decisions from late Sep 2026
- 280+** OEM manufacturer leads in outreach; first interested replies, expo meetings booked (MSPO Poland, Sep)
- £350K** UK DASA Open Call — defence-tech, planned
- €150K+** NATO DIANA / EU dual-use grants — co-bid with UA partner, planned

6-9 months from close to working MVP: GPS-free flight validated, ≤ 5 m CEP target, 10-unit OEM trial fleet.

Defence-nav M&A is funding the path to exit.

Active M&A and capital flows in defence-tech navigation.

01

Strategic acquisition by drone prime

Anduril, Shield AI, Northrop, Rheinmetall — actively acquiring nav stacks.

Auterion \$130M Series B · ~\$600M val

02

Acquisition by avionics / defence integrator

Garmin, Honeywell, BAE, Leonardo — building GPS-denied product lines.

Xona \$320M raised to date · Asio \$15M + Anduril partnership

03

Series A → B → IPO path

If beachhead converts to \$20M+ ARR by Y4 — defence-tech IPO window.

Anello \$25M + \$20M DoD APFIT

Nobody owns the map + honesty slot.

Landscape scan, Aug 2026 — nobody offers a sovereign fresh map with a transparent CEP.

Company	HQ	Funding / traction	Accuracy (denied)	Position vs ONS
Theseus	USA	\$4.3M seed · SOCOM	52 m median / 564 km	Closest direct comp — same approach, no own map, no confidence output
Vermeer	US-UA	\$10M Series A (Draper)	not disclosed	Combat-proven kit at ~\$20K/unit — not attritable-priced
Asio Tech	Israel	\$15M + Anduril · 10K combat hrs	drift-free, CEP n/d	Premium module benchmark — CEP & price not published
Twist Robotics	Ukraine	500K km · ~25K sorties	~20 m, no cumulative drift	Mass UA incumbent — we beat on accuracy target & fresh map
Delian	Greece	~\$20M	≤15 m, ≤0.1% error	Deep-strike focus — adjacent to our phase 2
ONS / FLY BY	UK-UA	\$500K Pre-Seed (this round)	17 m median field-tested · ≤5 m target	Sovereign fresh map IP + honest confidence + attritable price

Sources: open reporting Aug 2026 (Theseus field report, Brave1, Draper, company disclosures). Independent benchmarks are not ONS results.